

Abhishek Soundalgekar

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EDUCATION

University of Southern California, Masters in Computer Science | Los Angeles, California Jan 2025 – Dec 2026
Courses at USC: Artificial Intelligence | Machine Learning | Algorithms | Operating Systems | IR | NLP | DL

Pune Institute of Computer Technology, BTech. CS. & Honors degree in AI&ML | Pune, India June 2020 – July 2024

EXPERIENCE

Hewlett Packard Enterprise (HPE), Software Engineering Intern | San Jose, California May 2026 – Aug 2026

- Developing an ML-powered network traffic classifier for HPE Aruba Access Points, enabling real-time identification of Teams, AirPlay, bulk file transfers, and video streams across 4+ distinct application traffic classes.
- Engineered a high-performance C++ packet analytics pipeline from scratch with custom PCAP parsing, per-flow feature extraction, and model validation across 23 network telemetry features, producing labeled training datasets at scale.
- Designed a lightweight MLP classifier for edge deployment on constrained Aruba APs, achieving sub-millisecond inference and 95%+ accuracy; architected integration hooks for Aruba QoS and traffic-management systems infrastructure-wide.

USC Alzheimer's Therapeutic Research Institute, SWE Intern | Remote | San Diego, California Jan 2026 – May 2026

- Built 6 full-stack clinical research modules using Django, React, and PostgreSQL, serving 20+ active researchers across Agile sprints tracked in Jira, accelerating clinical trial data collection and workflows by 30%.
- Enforced Test-Driven Development (TDD) across all backend Django services, implementing unit, integration, and functional test suites achieving 84% code coverage and reducing post-deployment defects by 45% across 6 modules.
- Engineered MedDRA Adverse Event Coding platform features including role-based access controls, automated reporting dashboards, and REST API integrations for EHR systems, improving AE documentation speed for 25+ clinical staff.

University of Southern California, Software Developer, IT Systems | Los Angeles, California May 2025 – Present

- Engineered Python-based automation and provisioning workflows using anomaly detection techniques, reducing operational workload by 70% and account setup time by 80%.
- Developed predictive analytics and Generative AI applications to automate 100+ operational workflows, reducing ticket resolution time by 50% and improving scalability of technology support services.

AlgoAnalytics Private Limited, Data Scientist | Pune, India Aug 2023 – Dec 2023

- Engineered sentiment analysis pipelines using SVM, Naive Bayes, and Decision Tree models on 50k+ financial news articles, achieving 92% classification accuracy for large-scale market intelligence and forecasting applications.
- Built automated NER and feature engineering pipelines in Python, processing 5 GB of financial data daily, improving forecasting accuracy by 18% and reducing model training time by 35% for scalable production model training pipelines.

Sarvatra Technologies Private Limited, Software Developer Intern | Pune, India Aug 2023 – Jun 2024

- Built a secure UPI switch simulator using RSA-2048 encryption, emulating end-to-end digital payment workflows across bank and merchant nodes and sustaining 500+ simulated transactions per second under load testing environments.
- Optimized Flask and FastAPI microservices via async I/O and query tuning, improving system throughput by 30%; implemented Docker CI/CD with GitHub Actions across 6 services, cutting deploy time by 50% in production.

SKILLS & ACHIEVEMENTS

Languages	Python, C++, Java, SQL, TypeScript, Go
Concepts	OOP, System Design, Data Structures & Algorithms, Distributed Systems, Agile/Scrum, Scalability
Tools	LightGBM, Kubernetes, AWS (EC2, Lambda, SageMaker), GitHub Actions, Bash, React, Django
Systems/Backend	Linux, Git, Docker, REST APIs, Flask, FastAPI, Node.js, PostgreSQL, CI/CD
AI/ML	NumPy, Pandas, scikit-learn, PyTorch, TensorFlow, Reinforcement Learning, Transformers, Optimization
Achievements	Solved 500+ DSA problems on GFG (USC Rank 1); LeetCode: 500+ problems, 20+ badges, Institute Rank 1.

PROJECTS

Foundations of AI Mega-Project | CSCI-561, USC | C++, Python, TensorFlow, OpenAI Gym Spring 2025

- Applied genetic algorithms to solve 3D TSP, reaching 98.3% optimality across 188 peers under constrained compute.
- Built search agents for 5×5 Go and Pac-Man using minimax and alpha-beta pruning, ranking top in class AI tournament.
- Implemented Viterbi POMDP engine decoding hidden states over 20-step sequences with 100% accuracy under 10 ms/step.

Distributed Cloud File System | Personal Project | C++, Python, AWS S3, Docker, Kubernetes Ongoing

- Designed a distributed file system with sharding and replication, ensuring 99.99% availability and near-linear scaling.
- Implemented load balancing and S3-backed caching, cutting read/write latency by 45% and scaling to high concurrency.
- Integrated Docker and Kubernetes CI/CD to automate deployment, ensuring fault tolerance and high system throughput.

Analysis of Algorithms Final Project | CSCI-570, USC | C++, Python, NumPy, Matplotlib Spring 2025

- Optimized sequence alignment with dynamic programming on 100+ DNA/RNA pairs (20 kbp), achieving 0.08s runtime.
- Designed a memory-optimized DP variant reducing space complexity from $O(mn)$ to $O(n)$, cutting memory usage by 70%.
- Automated benchmarking across 10 dataset sizes from 1 to 20 Kb, validating runtime efficiency and alignment accuracy.